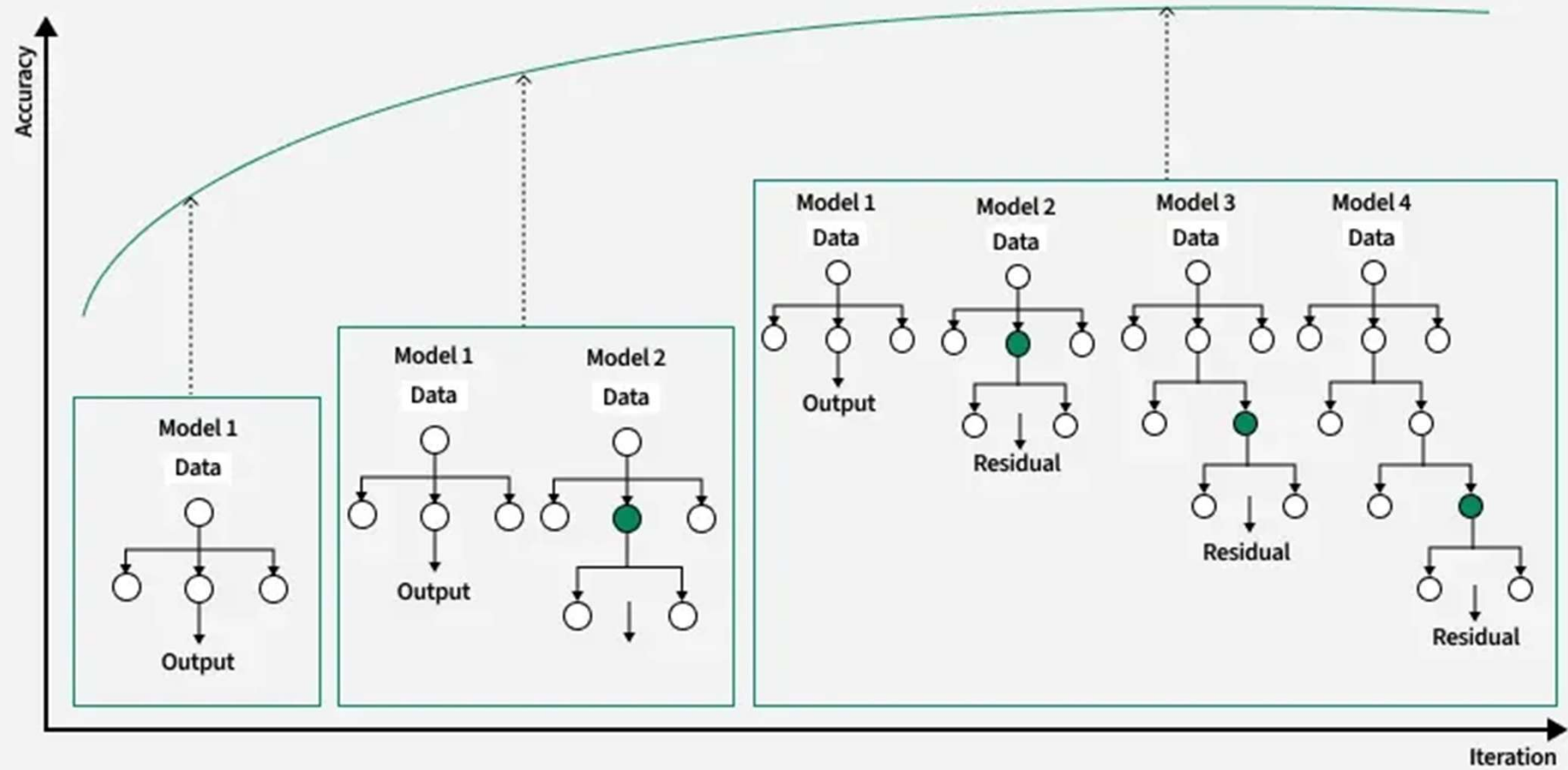


What is LightGBM?

- LightGBM stands for Light Gradient Boosting Machine.
- It is a fast and efficient machine learning algorithm.
- Used for classification and regression tasks.
- Built on decision tree boosting techniques.

How LightGBM Works

- Step 1: Train the first decision tree.
- Step 2: Calculate prediction errors.
- Step 3: Build new trees to reduce errors.
- Step 4: Grow trees leaf-wise for better accuracy.
- Step 5: Combine all trees for final prediction.



Advantages and Disadvantages

- Advantages:
 - - Faster than many boosting algorithms
 - - Handles large-scale data efficiently
- Disadvantages:
 - - Can overfit on small datasets
 - - More complex to tune parameters

Simple Real-Life Example

- Imagine a manager improving employee performance.
- After each review, weak areas are identified.
- Training focuses on improving those weak areas.
- LightGBM improves predictions similarly step by step.